PCB Axial-Flux Motor: Thermal Analysis Report

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Summary

With the rotor(s) held still and the bench fans on, the hottest board reaches 100 °C at a sustained torque of 47 mN·m for Rev0, 77 mN·m for Rev1-S and 111 mN·m for Rev1-D. Rev1-D therefore sustains 2.3× Rev0's torque, even though its current limit is lower (1.31 A_pk against 1.75 A_pk).

The study uses ANSYS Icepak 2025 R2 steady-state conjugate heat transfer (solid conduction plus natural and fan-forced convection of air) on each motor inside an assumed housing on the test bench. The copper loss $P_{Cu} = 3 \cdot I_{rms}^2 \cdot R(T)$ is applied to the winding region of every board. $R(T)$ is updated from the solved winding temperature until the loss changes by less than 1 %. A lumped thermal network, calibrated to the Rev0 Icepak results within ±7 %, adds heating curves and an estimate for a turning rotor.

Design	Magnets	Continuous I_pk	Continuous torque	Copper loss at 20 mN·m	Magnets at the limit
Rev0	16	1.75 A	47 mN·m	1.22 W	≈74 °C
Rev1-S	32	1.33 A	77 mN·m	0.52 W	≈74 °C
Rev1-D	48	1.31 A	111 mN·m	0.24 W	≈73 °C

All three designs are limited by the boards, with the N35 magnets close behind their 80 °C rating. The limits are conservative: they assume still rotors, an assumed housing and 30 °C hood air. For Rev0, the lumped model estimates that a rotor turning at 1000 rpm raises the continuous current from 1.77 A_pk to about 2.25 A_pk.

1. Scope and purpose

The practical current rating of this motor is a thermal limit, not a magnetic one. The electromagnetic study (separate report) showed torque rises linearly with current; this study finds the current, and therefore the torque, that each design can sustain without overheating.

The coreless PCB stator makes this the dominant limit. Each board carries six 35 µm copper layers in 1 mm of FR-4, and there is no iron stator to act as a heat sink. All heat must leave through the board faces, the thin air gaps and the housing.

The study answers the questions set for the thermal deliverable of AK-07:

- copper loss against phase current, including the rise of copper resistance with temperature;
- peak and average board temperature;
- temperature distribution through each five-board stack;
- temperature rise against current, and a defensible continuous-current limit;
- steady state, plus a heating curve for the time scale of a bench test;
- magnet temperature, because N35 is rated to about 80 °C.

Three designs are covered: Rev0 (one 5-board stator, one 16-magnet rotor), Rev1-S (two stators, two single-sided rotors, 32 magnets) and Rev1-D (two stators, a double-sided centre rotor plus an end rotor, 48 magnets).

2. Heat source: copper loss

Copper loss is the only heat source modelled. In a coreless stator at about 133 Hz (1000 rpm, 16 poles) the copper is thin compared with the skin depth (about 5.7 mm), so the DC resistance applies. There is no stator iron loss, and the rotor back-iron sees an almost steady field, so its loss is neglected.

For balanced sinusoidal phase currents:

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P_{\mathrm{Cu}} = 3 I_{\mathrm{rms}}^2 R_\phi(T) = \tfrac{3}{2}
I_{\mathrm{pk}}^2 R_\phi(T), \qquad R(T) = R_{20}\,[1 + 0.00393\,(T - 20)]
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The board-phase resistance is R20 = 0.2777 Ω (Maxwell Gate 1, mean of phases A, B and C). Each board therefore dissipates 1.5·I_pk²·R(T), and the boards of a stack carry the same current in series.

Design	Boards	R_φ at 20 °C	Loss at 1 A_pk, 20 °C	Loss at 2 A_pk, 20 °C	Loss at 2 A_pk, solved (fans on)
Rev0	5	1.389 Ω	2.08 W	8.33 W	11.6 W at 123 °C
Rev1-S	10	2.777 Ω	4.17 W	16.7 W	28.9 W at 220 °C
Rev1-D	10	2.777 Ω	4.17 W	16.7 W	29.8 W at 233 °C

The last column is why R(T) must be included. At high current the copper heats, its resistance rises, and the loss grows by 40 % (Rev0) to 80 % (Rev1) above the 20 °C value. Temperature rise therefore climbs faster than I².

3. Thermal model

Each motor is modelled in ANSYS Icepak 2025 R2 (steady state, laminar flow with gravity, conduction in every solid). The model is built by a PyAEDT 1.8 script, validated and frozen; separate scripts copy the frozen build and solve it, so geometry is never rebuilt between cases.

3.1 Geometry

- **Boards:** 12-sided, 93 mm across flats, 25 mm bore, 1.02032 mm pitch (1.0 mm board plus solder mask), stacked face to face. The copper loss is applied only to the winding annulus (r = 16–40 mm); the outer ring and bore carry no heat.
- **Rotors:** 16 N35 magnets of 20 × 6 × 2 mm at 28 mm radius, on an 86 mm steel_1010 disc 2 mm thick (19 mm bore), 1.0 mm air gap. Rev1-D's centre rotor is a 2 mm plate with magnets on both faces.
- **Shaft and bearings:** 19 mm steel shaft through both end plates; each bearing is an effective ring (k = 3.6 W/m·K, about 0.3 W/K).
- **Housing [assumed]:** based on the axial-flux test rig in the lab: acrylic tube, 120 mm inner diameter, 3 mm wall, 60 mm long, sealed; 140 × 140 × 6 mm aluminium end plates on an aluminium base; six M3 steel tie rods through six of the twelve 3.1 mm mounting holes (90 mm PCD).
- **Bench hood:** the computational region around the housing (±100 mm in x, +100 mm up, ±150 mm along the shaft). The bench side is an adiabatic wall.

Design	Stator S1 (mm)	Rotor R1	Stator S2 (mm)	Rotor R2
Rev0	20.000–25.102	magnets + 2 mm steel	—	—
Rev1-S	20.000–25.102	magnets + 2 mm steel	31.102–36.203	magnets + 2 mm steel, top 41.203
Rev1-D	20.000–25.102	magnets + shared 2 mm plate + magnets	33.102–38.203	magnets + 2 mm steel, top 43.203

Axial positions are measured from the inside face of the stator-side end plate. Rev1-D uses the axial levels of the frozen Maxwell Rev1-D build. Rev1-S assumes a 1 mm clearance between R1's bare steel and S2.

3.2 Materials

Part	Material	k (W/m·K)	ρ (kg/m³)	c_p (J/kg·K)
Boards	PCB effective, anisotropic	40.3 in-plane, 0.334 through	2600	900
Rotor steel, shaft, rods	steel_1010	52	7870	460
Magnets	NdFeB (N35)	9	7500	420
Housing tube	acrylic	0.19	1190	1470
End plates, base	aluminium 6061	167	2700	900

The board conductivity treats each board as 6 × 35 µm copper at 50 % coverage within 1.02 mm of FR-4 (copper fraction 10.3 %). In-plane, the layers act in parallel (0.103 × 385 + 0.897 × 0.8 = 40.3 W/m·K); through the thickness they act in series (0.334 W/m·K).

3.3 Boundary conditions

- Hood air at the motor: 30 °C [assumed; to be measured on the bench].
- Bench fans: 1 m/s along the shaft axis, entering through the −Z face [assumed]; the "fans off" cases set 0 m/s. All other region faces are pressure openings at 30 °C.
- Gravity along −Y (the motor's axis is horizontal).
- Radiation is not modelled (conservative; the lumped model puts it at about a quarter of the outside heat transfer).
- Rotors are **still**. This is the stalled, no-stirring case and gives conservative (hot) results; the effect of rotation is estimated with the lumped model (Section 5).

3.4 Mesh and solution

The global mesh uses an 8 mm maximum cell and a 0.25 mm minimum gap, refined to 2 × 2 × 0.5 mm over the stator stacks. Each steady solve takes about 2.5–3 minutes on the 16 GB lab PCs.

Each case starts with the loss at a guessed temperature, solves, reads the mean winding temperature of each board or stator, recomputes R(T) and the loss, and solves again. It stops when the loss changes by less than 1 %, usually after 2–5 solves. Temperatures are taken as winding means, winding maxima, magnet volume means and point monitors at mid-thickness of each board (r = 28 mm).

For Rev1, each five-board stator is one merged board body plus one winding annulus. With ten separate thin boards, Icepak's mesher dropped one winding regardless of mesh settings (Section 8). The boards touch with no contact resistance in either model, so merging does not change the physics.

4. Rev0 results (Icepak, rotor still)

At its realistic operating point (20 mN·m, 0.737 A_pk) Rev0's boards run at 41 °C, 11 K above the hood air. At 2 A_pk they reach 123 °C with the fans on and 133 °C with them off.

Case	Copper loss	Board winding means B1 → B5 (°C)	Hotspot	Magnets	Stack → hood
0.737 A_pk (20 mN·m), fans on	1.22 W	39.2 / 40.4 / 40.7 / 40.6 / 40.7	40.9 °C	36.8 °C	8.8 K/W
1 A_pk, fans on	2.32 W	47.2 / 49.3 / 49.9 / 49.8 / 49.9	50.3 °C	42.7 °C	8.6 K/W
2 A_pk, fans on	11.57 W	110 / 120 / 123 / 122 / 123	124.9 °C	89.2 °C	8.0 K/W
2 A_pk, fans off	11.87 W	119 / 130 / 133 / 132 / 133	134.5 °C	98.0 °C	8.6 K/W

B1 is the board next to the rotor. Magnet temperatures are the centre-point monitor (2 A fans off: volume means 95–98 °C over the 16 magnets).

Through the stack. Board 1 is the coolest by about 3 K at 1 A and 13 K at 2 A; boards 3–5 are the hottest and almost equal. Board 1 loses heat across the 1 mm gap to the rotor, while boards 3–5 can only lose heat through their neighbours and their thin rims. The hotspot is only 0.4–2 K above the hottest board mean, because the copper planes spread heat well in-plane (40 W/m·K).

Thermal resistance. The resistance from the stack to the hood air stays between 8.0 and 8.8 K/W across all cases. It falls slightly at higher loss because natural convection strengthens with temperature difference. The fans lower it by about 7 % (8.0 against 8.6 K/W at 2 A). They cool only the outside of the sealed housing, so most of the resistance is inside it.

Other parts at 2 A, fans off. Rotor steel 95.6 °C (mean), shaft 75.7 °C, bearings about 65 °C, end plates about 60 °C, acrylic shell 60.8 °C, base 59.7 °C. The housing therefore runs about 30 K above the hood air, and the remaining drop of about 70 K is between the stack and the housing.

Magnets. With the rotor still, the magnets reach about 60 % of the boards' temperature rise. They exchange heat with board 1 across the air gap, and the rotor has only weak paths out (cavity air and the shaft). This is why they sit close to their 80 °C rating at the board limit.

5. Rev0 heating curves and the effect of rotation (lumped model)

A calibrated lumped network predicts that a turning rotor raises Rev0's continuous current from 1.77 A_pk to about 2.25 A_pk, and that heat-up takes 10–25 minutes to reach two-thirds of the final rise.

5.1 Network and calibration

The network has twelve nodes: the five boards, rotor, shaft, tie rods, cavity air, acrylic shell and the two end plates, with the hood air as the fixed reference. Conductances come from geometry and standard correlations:

- board to board through the FR-4 and a 500 W/m²K mask contact;
- board 1 to rotor by conduction across the 1 mm (magnets) and 3 mm (steel) gaps;
- surfaces to cavity air by natural convection, plus stirring by the rotor when it turns (free rotating disc, Nu = 0.36·Re^0.5, half of it applied to the cavity);
- housing to hood air by forced convection (cylinder in cross-flow at the fan speed) and radiation.

Three still-rotor parameters were fitted to the four Icepak Rev0 cases: cavity natural convection 4.66 W/m²K, rotor–shaft conductance 19.3 W/K and bearing conductance 0.01 W/K. The fit reproduces Icepak's hottest board and magnet temperatures within ±7 % of their rise (2 A fans on: 126 °C and 92 °C against Icepak's 123 °C and 89 °C).

5.2 Steady results

Case (fans on)	Copper loss	Hottest board	Rotor / magnets	Time to 63 % of rise
20 mN·m (0.737 A_pk), 1000 rpm	1.2 W	36 °C	33 °C	9 min
1 A_pk, 1000 rpm	2.3 W	41 °C	35 °C	9 min
2 A_pk, 1000 rpm	10.3 W	82 °C	55 °C	11 min
2 A_pk, rotor locked	11.8 W	126 °C	92 °C	24 min

Continuous I_pk (board ⩽ 100 °C, magnets ⩽ 80 °C)	Fans on	Fans off
Rotor turning at 1000 rpm	2.25 A	2.13 A
Rotor locked	1.77 A	1.71 A

The locked-rotor limit agrees with Icepak (about 1.75–1.8 A_pk). The turning-rotor figures rest on the disc-stirring correlation, which is the model's largest uncertainty: halving or increasing the stirring by half moves the 2 A, 1000 rpm board temperature between 96 °C and 75 °C. Bench measurements will calibrate it.

5.3 Heating curves

A locked rotor at 2 A_pk is the case to watch on the bench: the hottest board passes 60 °C after about 6 minutes, 80 °C after about 15 minutes and 100 °C after about 37 minutes. A turning rotor at 2 A settles at about 82 °C.

6. Rev1 results (Icepak, rotors still)

Rev1-S and Rev1-D heat almost identically at equal current: at 1 A_pk their hottest boards reach 65.7 °C and 67.1 °C. They differ in torque per amp, not in thermal behaviour. Both carry twice Rev0's copper in the same housing, so at equal current they run much hotter than Rev0; at equal torque they run cooler (Section 7).

6.1 Rev1-D (48 magnets)

Case	I_pk	Copper loss	S1 boards B1 → B5 (°C)	S2 boards B1 → B5 (°C)	Hotspot	Magnets R1A / R1B / R2
20 mN·m, fans on	0.234 A	0.24 W	32.0 / 32.0 / 32.1 / 32.1 / 32.1	31.9 (all)	32.0 °C	31.2 / 31.2 / 31.1 °C
1 A, fans on	1.0 A	4.87 W	65.9 / 66.5 / 66.8 / 66.9 / 66.7	63.9 / 64.3 / 64.7 / 64.2 / 63.7	67.1 °C	52.6 / 52.6 / 50.6 °C
2 A, fans on	2.0 A	29.8 W	226 / 230 / 231 / 232 / 231	215 / 217 / 219 / 216 / 213	233.4 °C	153 / 153 / 143 °C
2 A, fans off	2.0 A	30.9 W	245 / 249 / 250 / 251 / 250	233 / 235 / 238 / 235 / 232	252.3 °C	170 / 170 / 159 °C

6.2 Rev1-S (32 magnets)

Case	I_pk	Copper loss	S1 boards B1 → B5 (°C)	S2 boards B1 → B5 (°C)	Hotspot	Magnets R1A / R2
20 mN·m, fans on	0.345 A	0.52 W	34.0 / 34.1 / 34.1 / 34.2 / 34.2	33.7 / 33.7 / 33.8 / 33.7 / 33.7	34.2 °C	32.6 / 32.3 °C
1 A, fans on	1.0 A	4.84 W	64.5 / 65.1 / 65.3 / 65.4 / 65.3	61.9 / 62.3 / 62.6 / 62.1 / 61.7	65.7 °C	52.5 / 49.9 °C
2 A, fans on	2.0 A	28.9 W	214 / 217 / 218 / 219 / 218	200 / 202 / 203 / 201 / 199	220.4 °C	150 / 136 °C

Case	I_pk	Copper loss	S1 boards B1 → B5 (°C)	S2 boards B1 → B5 (°C)	Hotspot	Magnets R1A / R2
2 A, fans off	2.0 A	29.9 W	230 / 233 / 234 / 235 / 234	215 / 217 / 219 / 217 / 214	236.8 °C	164 / 150 °C

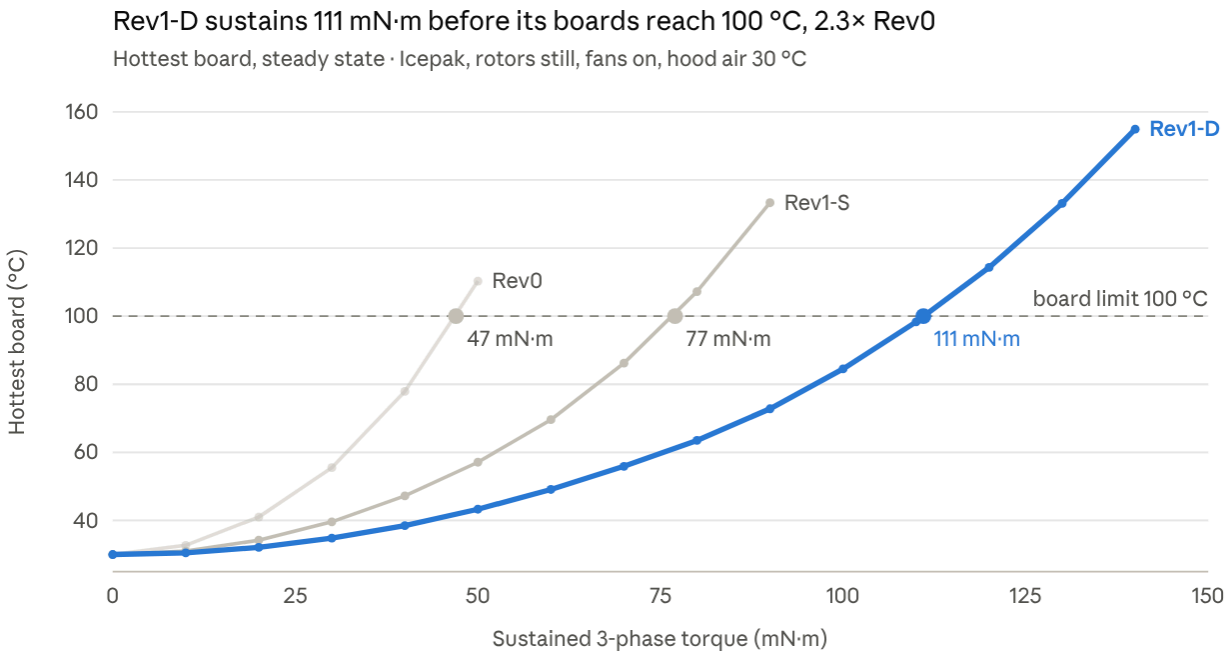
In both tables B1 is the board of each stator next to rotor R1. Board temperatures are point monitors at mid-thickness (r = 28 mm); the hotspot is the winding maximum.

6.3 What the Rev1 results show

- **The 20 mN·m point is trivial for both.** Rev1-D needs only 0.234 A_pk and rises 2 K; Rev1-S needs 0.345 A_pk and rises 4 K.
- **Stator S1 runs hotter than S2,** by 2–3 K at 1 A and up to 15 K at 2 A. S1 sits between the stator-side end plate (20 mm away) and the R1 rotor, which is itself heated by both stators. S2′s outer side faces R2, which is closer to the rotor-side end plate.
- **Rotor R1 is the hottest rotor.** In Rev1-D both its magnet layers (R1A, R1B) read the same, because the shared 2 mm plate conducts well between them.
- **2 A is far beyond Rev1′s capability.** Boards reach 213–252 °C and the magnets 136–170 °C, well past FR-4′s glass transition (about 130–140 °C), the solder mask and N35′s 80 °C rating. Copper resistance rises about 80–90 %, so the loss reaches 29–31 W against 16.7 W at 20 °C.
- **The stack-to-hood resistance is lower than Rev0′s** (about 6.8–7.6 K/W against 8.0–8.8 K/W), because two stators spread the heat over more surface. It is not halved, though, because both stators share one sealed housing.

7. Design comparison: continuous torque

Compared at equal torque rather than equal current, Rev1-D runs coolest: it sustains 111 mN·m at the 100 °C board limit, against 77 mN·m for Rev1-S and 47 mN·m for Rev0.



Icepak steady results (three converged fans-on cases per design), interpolated with copper R(T) · 2026-10-08

Each curve is built from that design's three converged fans-on Icepak cases. The stack-to-hood thermal resistance is interpolated against loss, and the loss is solved with R(T) at each torque (torque = 1.5 × stack constant K × I_pk, with K = 18.02, 38.48 and 56.68 mN·m/A for Rev0, Rev1-S and Rev1-D).

Why Rev1 wins despite a lower current limit. Doubling the boards doubles the copper loss at a given current, but the Rev1 designs give 2.1× (Rev1-S) and 3.1× (Rev1-D) Rev0's torque per amp. At equal torque their current is lower by that factor, and loss falls with its square. The extra stator surface also lowers the thermal resistance by about 15 %.

Rev1-D against Rev1-S. Thermally they are the same machine (Section 6), so Rev1-D's 1.45× sustained-torque advantage comes entirely from its 1.47× higher torque constant. That costs 16 more magnets.

Magnet margin. At each design's board limit the magnets reach 73–74 °C, inside N35's 80 °C rating but with little margin. If the real housing runs hotter, or if magnet heating proves the tighter limit on the bench, N38SH or similar high-temperature grades would remove the concern.

8. Verification and issues found

The reported results passed four checks; five problems found during the work were fixed, and the affected cases were re-solved.

8.1 Checks on the results

Check	Result
Energy scaling: temperature rise per watt constant across cases	Rev0 8.0–8.8 K/W, Rev1 6.6–7.6 K/W; varies only through natural convection
Convergence of the R(T) loop	final loss change 0.2–1.1 % per case (target < 1 %); 2–5 solves each
Two independent models agree (still rotor)	lumped network within ±7 % of Icepak's temperature rise for Rev0
Physical trends	board next to the rotor coolest; fans cool by 7–10 K at 2 A; equal-current Rev1-S and Rev1-D within 1.5 K

8.2 Issues found and fixed

Issue	Effect	Fix
Icepak designs come with a default region	the hood region could not be created	the build deletes the default region first
PyAEDT 1.8 gravity codes 0/1/2 mean −X/−Y/−Z	first draft would have pointed gravity up (no error raised)	gravity set along −Y, checked against the PyAEDT source
"Maximum" of a thin magnet's volume reads mesh nodes shared with the hot air gap	Rev0 log showed magnets at ≈128 °C (real ≈96 °C)	magnet temperature taken as the volume mean, cross-checked with a centre point monitor
Ten separate thin boards per Rev1 motor	the mesher left one winding (and with other settings, other parts) without mesh, so its heat was missing	each stator merged into one board body plus one winding annulus; the build now meshes and refuses to freeze if any part has no mesh
Temperatures above 200 assumed to be Kelvin	Rev1-D at 2 A read as −44 °C, so its R(T) loop could not converge	conversion removed (the field calculator returns °C); invalid cases re-solved automatically

No reported number comes from a run affected by these issues.

9. Assumptions and limitations

The absolute temperatures depend most on the assumed housing and on the rotors being still; the comparison between designs is much less sensitive, because all three share the same assumptions.

Assumption or limitation	Direction of error	Effect and how to resolve it
Housing geometry and materials assumed (real housing not yet built)	either	Lumped model: an aluminium instead of acrylic shell changes the boards by under 1 K, because the resistance is inside the housing. A vented housing lowers the 2 A board temperature by about 5 K. Rebuild with real dimensions when known.
Rotors still in Icepak	conservative (hot)	Rotation stirs the cavity air. Lumped model: Rev0 limit rises from 1.77 to about 2.25 A_pk at 1000 rpm, but this depends on an uncalibrated correlation. Measure on the bench.
Hood air 30 °C, fans 1 m/s	either	Limits scale with the allowed rise: every 1 K hotter air removes about 1.4 % of the 70 K rise. Measure the air at the motor.
No radiation	conservative	About a quarter of the housing's outside heat transfer, so modest.
Board-to-board contact perfect in Icepak	slightly optimistic	Lumped model: a 100–2000 W/m²K contact range moves the 2 A board temperature by about 3 K.
Copper as a uniform layer (50 % coverage, anisotropic k)	small	Hotspots within a track are not resolved; the in-plane spread keeps hotspot to mean within 2 K.
Copper loss only (no eddy, rotor or bearing loss)	slightly optimistic	Small at 133 Hz in 35 µm copper; rotor back-iron loss small at a near-steady field.
Rev1 stators merged into one body each	none expected	Same material and contact as the separate boards; per-board values from point monitors.

Assumption or limitation	Direction of error	Effect and how to resolve it
Rev1-S clearance R1 steel to S2 of 1 mm	either	Affects only S2′s coupling to R1; set from the mechanical design.
Limits for 100 °C board and 80 °C magnets	policy	A 105 °C board or N35 80 °C demagnetisation margin can be revised; the curves in Section 7 give any other limit directly.

10. Recommendations

Run the bench test inside these limits, and use it to calibrate the two numbers the model cannot fix on its own: the housing's real thermal resistance and the cooling from a turning rotor.

10.1 Operating limits for testing

Design	Continuous (rotor still)	Short-term at 2 A_pk, rotor locked	At 20 mN·m
Rev0	≤ 1.75 A_pk (47 mN·m)	a few minutes; 60 °C after 6 min, 100 °C after 37 min	0.74 A_pk, 41 °C
Rev1-S	≤ 1.33 A_pk (77 mN·m)	avoid; steady state 220 °C	0.35 A_pk, 34 °C
Rev1-D	≤ 1.31 A_pk (111 mN·m)	avoid; steady state 233 °C	0.23 A_pk, 32 °C

Set the drive's current limit to the continuous value until bench data shows a turning rotor allows more.

10.2 Measurements to take

1. Thermocouples on board 1 (next to the rotor) and board 5 (stator-side end), at the winding radius (about 28 mm).
2. One thermocouple in the hood air close to the housing, the model's boundary condition.
3. Winding resistance before and after each run, which gives the average copper temperature directly: $\Delta T = (R/R_0 - 1)/0.00393$.
4. An infrared reading of the magnets immediately after stopping.

5. A heat-up run at a fixed current (for example 1 A_pk, rotor locked) for at least 30 minutes, logging every minute, to fit the time constant.
6. The same run at 1000 rpm: the difference from step 5 calibrates the rotor-stirring term.

10.3 Model next steps

- Replace the housing defaults with the real housing once it is designed, and re-run (each design is under an hour on a lab PC).
- Calibrate the lumped model's stirring term with the bench data, then publish turning-rotor limits for all three designs.
- Optional: an Icepak transient to cross-check the heating curves, and a rotating-wall or moving-reference-frame model of the rotor if a supported option is found in AEDT 2025 R2.

Appendix: scripts, files and run record

Scripts are in OneDrive \EEE4022S Ansys files\Thermal and run from a Command Prompt with Python 3.13 and PyAEDT 1.8 against AEDT 2025 R2. AEDT works in C:\Users\Student\Rev0_work\thermal (or D:\New folder\Motor run\thermal where a D: drive exists); logs, CSV files and frozen builds are copied to Thermal\results. Each script starts its own AEDT session and closes it when finished.

Script	Role
REV0_ICEPAK_BUILD.py	builds, validates and freezes the Rev0 model (REV0_ICEPAK_FROZEN.aedt)
run_REV0_ICEPAK_STEADY.py	solves the four Rev0 cases with the R(T) loop; resumes after a crash
check_REV0_ICEPAK_TEMPS.py	read-only per-object temperatures and monitor values
REV1_ICEPAK_BUILD.py Rev1D / Rev1S	builds a Rev1 model (merged stators), meshes it, refuses to freeze if any part has no mesh
run_REV1_ICEPAK_STEADY.py Rev1D / Rev1S	solves the four Rev1 cases; re-solves any case holding invalid values
dump_REV1_MONITORS.py Rev1D / Rev1S	exports all point-monitor values (per-board temperatures) to JSON
rev0_thermal_lumped.py	calibrated lumped network: heating curves, turning-rotor estimate, current limits

Run (2026-10-08)	Cases	Finished
REV0_ICEPAK_STEADY_RUN1	0.737 A, 1 A, 2 A fans on; 2 A fans off	05:28
REV1D_ICEPAK_STEADY_RUN1	0.234 A, 1 A, 2 A fans on; 2 A fans off (2 A re-solved)	08:36
REV1S_ICEPAK_STEADY_RUN1	0.345 A, 1 A, 2 A fans on; 2 A fans off	09:2x

The electrical inputs come from the electromagnetic study: board-phase resistance 0.2777 Ω (Maxwell Gate 1) and stack torque constants 18.02, 38.48 and 56.68 mN·m/A (board-by-board FEM, stack sum and MATLAB model).